Finger Tracking Using Wrist-Worn EMG Sensors

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***Abstract*—This paper introduces WETrak, a finger tracking system using wrist-worn electromyography (EMG) sensors. Recent finger tracking methods mainly employ EMGs on armbands. Compared to a range of contactless methods using cameras or wireless, they are not limited by high computational costs, privacy concerns, and mobility, while unlike other wearable-based approaches, they do not require the deployment of sensors on the user's hands. However, users need to wear an additional armband on their forearm each time solely for tracking purpose, which hinders the widespread adoption of finger tracking in practice. This paper investigates the feasibility of moving EMG sensors from the forearm to the wrist for finger tracking. WETrak inherits the advantages of existing EMG-based armband tracking while avoiding the limitation of requiring additional armbands, which brings a strong incentive for integrating EMG sensors into wrist-worn wearables in the future. As sensor placement varies, we find new challenges in determing good locations to place sensors to gather useful information to capture all finger movements and using low-quality signals to still ensure accurate tracking. In this paper, we introduce new, efficient solutions to these problems. We develop a prototype, and the results show that WETrak outperforms the state-of-the-art method and performs consistently well under various settings.**

***Index Terms*—EMG sensors, finger tracking, mobile sensing, muscle distribution.**

I. Introduction

FINGER tracking is a technique for reconstructing the user's hand pose consisting of finger skeleton points over time, which can enable many promising applications in the fields of human-computer interaction (HCI) [1], [2], [3], [4], smart healthcare [5], [6], [7], robotics [8], etc. In the literature, many contactless methods employing cameras [9], [10], depth sensors [11], and wireless sensing signals [12], [13], [14], but they have inherent limitations such as high computational cost, privacy concerns, sensitivity to external factors, and/or limitations in mobility [15]. Therefore, wearable-based approaches have attracted considerable attention recently (see Section V for details), where representative designs employ electromyography (EMG) sensors [15], [16] to sense muscle electrical activity to track the movements of fingers.

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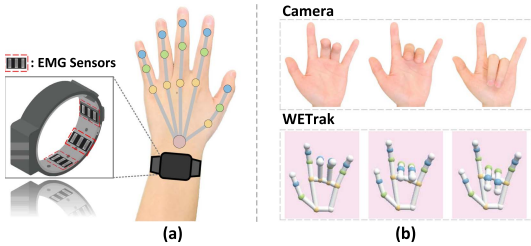

Fig. 1. Illustration of the WETrak design. (a) A conceptualized wrist-worn device equipped with EMG sensors that can perform finger tracking using WETrak. (b) A scene where the user makes a "love you" pose, with three snapshots of that pose from the camera and our tracked results, respectively.

State-of-the-art EMG-based tracking methods [15], [16] can overcome the aforementioned limitations of contactless methods and still achieve fine-grained finger tracking. In addition, they can bring the great convenience of requiring to wear no sensors on the user's hand. To provide a portable solution, recent EMG-based methods [15], [16] employ specific armbands, such as Myo [17], worn on the user's forearm. However, this additional cost that users need to purchase an extra armband only for tracking purpose and wear it on their forearm every time makes their tracking solution still less user-friendly and hinders the widespread adoption of finger tracking technology among the public.

In this paper, we observe that wrist-worn wearables such as smartwatches and wristbands are already popular [18], [19], [20]. Therefore, our main idea is to move the EMG sensor from the forearm to the wrist, while our goal is still to track fine-grained 3D hand poses, where Fig. 1(a) shows all 15 skeleton points of the five fingers that we aim to track in this paper. If feasible, we can foresee a strong incentive to integrate EMG sensors into wrist-worn wearables in the future, such as on the back of the device and its strap, as shown in Fig. 1(a). Wearable devices can thus inherit all the advantages of existing EMG-based armband tracking designs, while avoiding the limitations of requiring such an additional device. However, if the EMG sensors are directly moved from the forearm to the wrist, while employing the previous tracking methods [16], the tracking error would be substantial, e.g., exceed 10° (see details in Section IV-A). This indicates that designing an EMG tracking system based on the wrist necessitates overcoming several new challenges as follows.

1) Is it feasible to place EMG sensors at the wrist to track fine-grained 3D hand poses? The cross-sectional area of the

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forearm is much larger than that of the wrist. We find that the EMG sensor on the forearm can perceive more muscles related to hand movement (Section III-A), and these muscles are well developed and large in size, which is very convenient for sensing. On the contrary, the muscles at the wrist are relatively small and cover more of each other, making it difficult to deploy sensors. Therefore, it is currently unknown how to perform appropriate sensor deployment and efficient sensing information acquisition to track fingers at the wrist.

2) Low-quality sensing signals: Even if sensing at the wrist is feasible, accurate finger tracking is still difficult due to the relatively low quality of the sensing signal. The bioelectrical pathway is generated from the brain and transmitted to the hand. The electrical signals that travel across our forearm are generated by the muscles in the forearm section. Therefore, the signal strength collected by the armband is stronger. After the signal is transmitted to the wrist, it will experience significant attenuation [21], resulting in a greatly reduced signal strength and a low signal-to-noise ratio. Thus, we need to effectively extract useful sensing information from such degraded signals.

To address these challenges, we first perform a feasibility study and find that while there are far fewer muscles that can be sensed at the wrist, we can still find enough ones to cover the motion of all five fingers. Fortunately, these muscles can be sensed from the surface of wrist when the placement of the EMG sensors is aligned well with the location of these muscles. On the other hand, our study also reveals that the movement of each finger is most associated with different sets of muscles and presents different signal characteristics (Section III-A). Therefore, we design an effective neural network-based tracker with three specialized branches that mine the relationship between finger movements and the low-quality EMG signals driven by the associated muscles at the wrist (Section III-B). In addition, as finger tracker explores the mapping from EMG to finger movements via regression, we propose a novel approach to dynamically determine the appropriate search space to perform regression on different inputs. This method can efficiently reduce each search space and improve the reliability of the results obtained from weak signals (Section III-C).

In this paper, we present this wrist-worn EMG-based finger tracking design, called WETrak. We develop a prototype system with five EMG sensors and deploy the tracker on a Google Pixel 7 phone. Fig. 1(b) visualizes some tracking results of WETrak. To comprehensively evaluate its performance, we recruit 15 users in our experiments. Compared to the state-of-the-art EMG-based finger tracking method NeuroPose [16] using the Myo armband [17], our effective sensor placement and network design ensure that the tracking performance of WETrak is comparable to that of NeuroPose sensed on the forearm, while the final version of WETrak with adaptive regression can even outperform NeuroPose by 32.22%, with a tracking error of only 4.94–7.68°. The evaluation also shows that WETrak is robust under various settings and lightweight, outputting 500 frames of hand poses with 15 finger skeleton points per second. Since WETrak has great potential to enable many useful applications [22], [23], [24], [25], we will open source all code to facilitate future research in this area. In summary, this paper makes the following contributions:

- We reveal the feasibility of moving EMG sensors from the forearm to the wrist for finger tracking. It brings strong incentives to integrate EMG sensors into wrist-worn wearables and give them such tracking capabilities.
- We identify unique challenges when placing EMG sensors on the wrist for finger tracking, perform an insightful feasibility study to motivate our design, and propose novel solutions to address these challenges in WETrak.
- We develop a prototype system of WETrak and conduct extensive evaluations. Results show that WETrak outperforms the state-of-the-art method and consistently performs well across a wide range of system settings.

II. Background

A. Application Scenarios

With the ability to track fine-grained finger movements via wrist-worn sensors, WETrak has great potential to benefit or advance a variety of applications, including at least:

1) *Novel HCIs:* When EMG sensors are integrated into a smartwatch and its strap, we can endow it with a virtual hand operated by the user, so that the user can directly perform complex interactions with the smartwatch or other connected devices. On the other hand, when the hand moves out of the field of view of a VR/AR device's camera [3], [4], WETrak can also provide supplementary finger information as input.
2) *Healthcare:* Finger tracking plays a vital role in physical rehabilitation, especially for patients with hand injuries [5], [6], [7]. WETrak allows therapists to tailor treatment plans and effectively monitor recovery progress by meticulously tracking finger movement progress for better recovery.
3) *Robotics:* WETrak can enable remote control of robotic hands, especially in teleoperation scenarios involving dangerous or inaccessible settings [8], [26], [27]. Using finger tracking, teleoperators can increase the precision of robot movements, improving operational safety in challenging environments.

B. Finger Skeleton Model

As shown in Fig. 2(a), our hand contains 15 main skeleton points: five distal interphalangeal (**DIP**) joints, five proximal interphalangeal (**PIP**) joints, and five metacarpophalangeal (**MCP**) joints. Before discussing how to track their movements, we first introduce the coordinate system used in WETrak.

1) Coordinate system: We define the "X-Y" plane to be aligned with the palm, and "+Z" is the direction away from the palm, as shown in Fig. 2(b). The origin of the coordinate system is set at the center of the palm. The "+Y" direction is along the middle finger when the hand is naturally straight, and the "+X" direction is pointing to the little finger side.

2) Degrees of freedom (DoFs): On the other hand, there are two types of movement DoFs for finger joints:

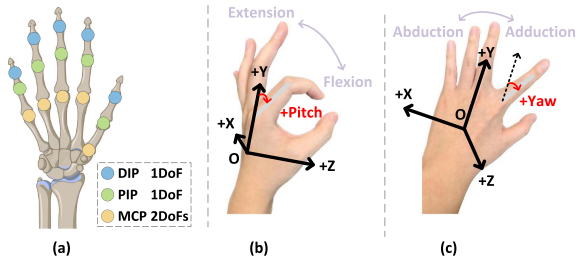

Fig. 2. (a) DoFs for all finger joints to track in WETrak. Illustration of (b) the pitch angle and (c) the yaw angle in the coordinate system used in WETrak.

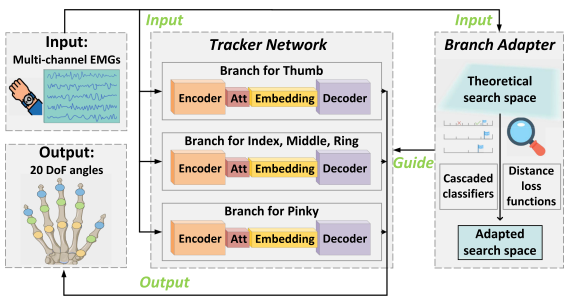

Fig. 3. Overview of the WETrak design.

- *Pitch:* When a finger joint is extended or flexed, the line segment between that joint and the next joint (towards the fingertip)[1] forms an angle with respect to the "Y" axis. The projection of this angle on the "Y-Z" plane is the pitch angle. Fig. 2(b) shows the pitch angle of the MCP joint of the index finger.
- *Yaw:* When a finger joint performs abduction or adduction along the "X-Y" plane, the angle (from the "Y" axis to the line segment between that joint and the next joint towards the fingertip) projected on the "X-Y" plane is the yaw angle. Fig. 2(c) shows an example for an MCP joint.

According to the kinematics of human hand motion [28], DIP and PIP joints have only one pitch DoF angle, while MCP joints have pitch and yaw two DoFs, as summarized in Fig. 2(a). Similar to the existing finger tracking methods [16], we also aim to track all these 20 DoF angles in WETrak,[2] which can be used to reconstruct hand poses by using a default or measured set of lengths between finger joints [16].

III. System Design

Fig. 3 shows the framework of WETrak, which consists of two main modules, the *tracker network* and the *branch adapter*.

1) Tracker network: The main responsibility of the tracker network is to infer all target DoFs from input EMG signals. As fingers can be divided into three groups (see reasons in Section III-A), the tracker network is designed with three branches, each responsible for one group, to better capture its signal-to-movement mapping features. In addition, we design a specialized spatial attention block to further improve the tracking accuracy.

[1]For each DIP joint, we use the line segment between it and the fingertip.

[2]The thumb's PIP may also have a slight yaw angle, which is the angle between the "Y" axis and the line segment formed by the thumb's PIP and DIP. Since this angle is very small compared to the yaw angle of the thumb MCP [29], we skip it in the current WETrak.

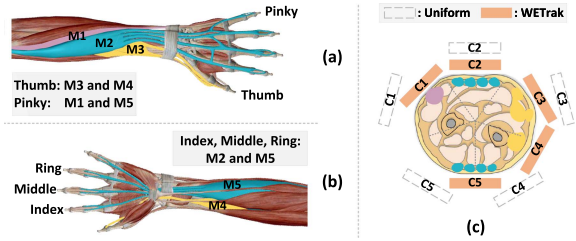

Fig. 4. Muscle distribution of the (a) front and (b) back of the forearm. (c) Our placement location of EMG sensors at the wrist.

2) Branch adapter: For each branch of the network, it essentially explores the mapping from EMG signals to finger movements via regression. We find that the search space for each regression is important for the performance and thus design a novel adapter in this module to dynamically adjust the appropriate search space to perform each regression on different inputs. When the adapter is enabled, it first determines the regression range before the network outputs the DoF results.

A. Feasibility of Finger Tracking With Wrist-Worn EMGs

Firstly, we should investigate the feasibility of performing finger tracking using wrist-worn EMG sensors. Fig. 4(a)-(b) depicts the distribution of muscles on the front and back of our forearm. We can clearly see that the cross-sectional area of the forearm is much larger than the wrist, and thus more muscles that control the movement of the hand can be seen in the forearm. At the same time, these muscles are large in size for easy perception. Therefore, the armband is designed to sense this area. In contrast, the muscles at the wrist overlap more, and these muscles are smaller, which is not conducive to deploying sensors for sensing.

1) Observations: Although the amount of muscles in the wrist is smaller, we have carefully studied the function of each muscle on the user's arm and found that the movements of all our fingers are completely covered by five muscles. Fortunately, these muscles can be sensed from the surface of the wrist. They are the extensor digiti minimi (M1), the extensor digitorum (M2), the pollicis muscle group (M3),[3] the flexor pollicis longus (M4), and the flexor digitorum superficialis (M5), as shown in Fig. 4(a)-(b). On the other hand, we also observed that each finger is mainly associated with two of M1–M5, as follows:

- **Thumb**: M3 and M4. **Pinky**: M1 and M5.
- **Index**, **middle**, and **ring** fingers: M2 and M5.

To experimentally understand these observations, we use five EMG sensors, denoted as channels C1 to C5, placed on the muscles of M1 to M5, respectively, and then we perform three feasibility study trials. Fig. 5 shows the results.

2) Results: When the user moves only the thumb, we can see that channels C3 and C4 exhibit dominant responses. Since the stretching and contracting of the muscles are not completely isolated, the movement of each finger also activates other

[3]M3 in Fig. 4(a) consists of three nearby muscles, the abductor pollicis longus, extensor pollicis brevis, and extensor pollicis longus.

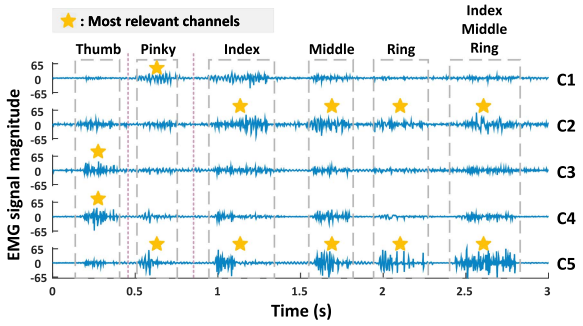

Fig. 5. EMG signals in channels C1–C5 when each finger moves. Stars represent the most relevant channels for the movements of each finger.

muscles, thus generating certain signals. Likewise, when only the pinky moves, C1 and C5 show the strongest signals, while the other three channels also show some response. Finally, when the index, middle or ring finger (or a combination of them) is moved, the main responses come from channels C2 and C5. In summary, this feasibility experiment provides three important insights to guide our design in the next subsection.

- *Sensor placement:* Unlike uniform placement on the armband, EMG sensors should be aligned with the location of the muscles M1–M5 at the wrist. Fig. 4(c) shows our deployment location, where we set C2 as the anchor point. C1 and C3-C5 should be closer to C2, offsets by 10°, 8°, 28° and 36° from a uniformly placed position, respectively.
- *Finger groups:* Based on the relationship of each finger to its most relevant muscles, the five fingers can be divided into three groups, with the thumb and pinky each as one group, and the remaining three as a third group. Since the characteristics of EMG signals mapped to corresponding finger movements vary within different groups, we will later leverage this property to guide our tracking design.
- *Sensing signals:* Since the stretching or contracting of muscles is not completely isolated, when a finger is moved, the most relevant muscles will have strong responses, and other muscles usually also generate certain signals. We will use this opportunity to further improve performance.

Armed with these useful observations, we now present the design of WETrak in the next subsection.

B. Design of Tracker Network

1) Network input and output: The tracker network takes EMG signals as input and outputs all estimated DoF angles.

a) Input and preprocessing: The input EMG signal is first passed through filters to retain the signal in the frequency range of 20–150 Hz, and remove the interference of other frequency bands [30]. The sampling rate of our EMG sensor is 500 Hz. Hence, 500 frames of EMG data will be generated per second. We denote each frame as $F_t^{in} = \{e_t^i\}_{i=1}^5$, where e_t^i is the EMG data sensed by channel i at time t, and then organize each input of the network as a window, containing s consecutive frames of EMG data. Therefore, the input to the network can be written as $\{F_t^{in}\}_{t=1}^s$. In Section IV, s is empirically set to 1000, corresponding to an input window size of two seconds.

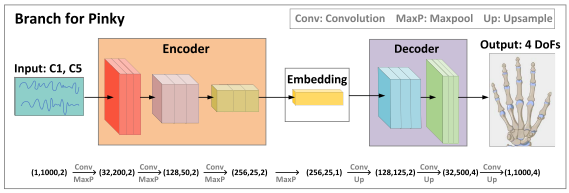

Fig. 6. Illustration of the branch backbone design for pinky, where the numbers in the parenthesis represent the dimension of each convolution layer.

b) Output: For s frames of EMG data within the input window, the network outputs s frames of output. Each output frame $F_t^{out} = \{o_t^j\}_{j=1}^{20}$ is all 20 estimated DoFs corresponding to its input frame (a snapshot of the hand poses), where each o_t^j is an estimated DoF at time t. Therefore, the tracker network aims to explore the following relation:

$$\{F_t^{in}\}_{t=1}^s \rightarrow \{F_t^{out}\}_{t=1}^s. \tag{1}$$

After the current input window finishes estimating DoFs, the window waits for a short period of time to load new data and perform the next estimation (i.e., sliding window). If the new window after sliding overlaps the window before sliding, we can simply discard the results already output in the previous window, and only keep the tracking results in the non-overlapping part in the new window. In Section IV-C, we investigate how the sliding step affects system energy efficiency.

2) Network structure: As revealed in Section III-A, the fingers can be divided into three groups, each group being more associated with two different muscles. Thus, our tracker network is designed as three branches, each responsible for one group, to better capture its signal-to-movement characteristics.

a) Branch backbone: We adopt an encoder-decoder architecture to design the backbone of each branch because of its recent great success in processing EMG signals [16]. However, the encoder-decoder architecture is just a framework. We customize it in WETrak. Next, we take "pinky" as an example to clearly introduce the structure of the branch backbone.

- *Encoder block:* This block performs a series of convolution operations (as shown in Fig. 6) from two most associated channels, C1 and C5, to learn the characteristics of the EMG signal, expressed as $\{f_t^{enc}\}_{t=1}^s$. These features encapsulate the spatial features between EMG signals and the temporal features within each channel.
- *Embedding block:* This block refines the features $\{f_t^{enc}\}_{t=1}^s$ extracted by the encoder to enhance feature quality and robustness before passing to the decoder, denoted as $\{f_t^{emb}\}_{t=1}^s$. We employ five residual layers as bridging components between the encoder and decoder, which can enhance the extracted features and overcome the vanishing gradient problem [31].
- *Decoder block:* This block uses the features $\{f_t^{emb}\}_{t=1}^s$ learned by the previous two blocks to reconstruct the four DoFs of pinky. By referencing the labels of these DoFs during training, the parameters of each branch can be updated for efficient feature extraction and accurate DoF estimation.

In summary, given all EMG inputs, each branch estimates all DoF angles for the fingers that this branch is responsible for. Then, all these 20 DoFs from different branches together form the overall output of the tracker network.

b) Spatial attention block: Before proceeding, we discuss a question about the input of each branch. Since each branch is responsible for a group of fingers, a natural design choice is to feed EMG signals from the two most relevant channels to each branch. As shown in our evaluation, this choice is feasible, but there is still room for improvement in performance.

In Section III-A, we observed that when a finger moves, besides its two most relevant channels, other channels also provide some useful information. Therefore, an alternative would be to send the EMG signals of all five channels to each branch so that no useful sensing information is lost. We find this solution helpful, but it causes another problem. For example, using EMG signals from all five channels, the performance of the "pinky" branch can indeed be improved when the pinky moves. However, when the pinky is stationary and the index finger is moving, this branch can still receive a considerable amount of signal from all channels, especially from C5. These signals can mislead the output of "pinky" branch.

Therefore, we need a mechanism to dynamically suppress EMG signals of each branch when its corresponding finger is not moving. To this end, we exploit the attention mechanism [32] and introduce a spatial attention block in our tracker network, as shown in Fig. 3. Without changing the network structure of each branch, the attention block only adds weights to all input channels of each branch, and it learns how to dynamically adjust the weights to pass or suppress the signals of each channel. Specifically, for each branch, the attention block multiplies a set of weights $W = \{w^i\}_{i=1}^5$ to the features $\{f_t^{enc}\}_{t=1}^s$ obtained from all five input channels $\{F_t^{in}\}_{t=1}^s$ as:[4]

$$f_t^{enc} \otimes W = \left\{ f_t^{enc,i} \times w^i \right\}_{i=1}^5, \text{ where } t = 1, \ldots, s,$$

where $f_t^{enc,i}$ is the feature extracted by the encoder corresponding to input channel i. W is updated by $W = g(\{f_t^{enc}\}_{t=1}^s)$, where $g(\cdot)$ is a lightweight convolution layer. The attention block adds weights to each branch, and different branches use different weights. As shown in Section IV, the attention block can effectively reduce the tracking error by 9.4%.

3) Loss function: To train the tracker network, we design two loss functions in WETrak. The first loss L_{mse} measures the squared difference between the estimated DoF o_t^j and the labeled value $\hat{o}_t^j$ (e.g., from Leap Motion [33]):

$$L_{mse} = \frac{1}{s} \sum_{t=1}^{s} \frac{1}{20} \sum_{j=1}^{20} \left(o_t^j - \hat{o}_t^j \right)^2, \tag{2}$$

where s is the number of data frames in the input window. The second loss L_{sim} takes into account the temporal continuity of finger movements, which computes the cosine similarity between the estimated movement trajectory and labeled one:

$$L_{sim} = 1 - \frac{1}{s} \sum_{t=1}^{s} \frac{1}{20} \frac{\sum_{j=1}^{20} o_t^j \times \hat{o}_t^j}{\left\| \sum_{j=1}^{20} o_t^j \right\| \times \left\| \sum_{j=1}^{20} \hat{o}_t^j \right\|}, \tag{3}$$

where $\| \cdot \|$ represents the L2 norm. The overall loss $L_{mse} + L_{sim}$ characterizes the regression error between predicted DoFs and labeled DoFs, and we minimize this loss during training.

C. Design of Branch Adapter

As revealed in our evaluation, the tracker network above, guided by our study of sensor placement on the wrist, can achieve comparable performance to the state-of-the-art method using armbands [16], which have better EMG placement. However, we find that the second branch in Fig. 3, dealing with the index, middle and ring fingers, has a relatively larger error than the other two branches because it uses one branch to track 12 DoFs of three fingers, which is more complicated. A naive solution would be to further expand this branch into three sub-branches. However, the future positioning of WETrak is to be integrated into wearable devices, and its computation will be performed by the phone or even directly by the wearable device. Thus, we tend to design with as few branches as possible to avoid unnecessary energy and memory overhead.

Opportunity: In WETrak, we identify opportunities to further improve the tracking performance of this branch without increasing the complexity of the network structure. Specifically, the branch network is essentially a regression model that maps EMG signals to DoFs. In our current design, we regress each DoF at its theoretical maximum based on the kinematics study [34]. However, we observe that not every finger moves at every moment, and even if they do move, they may not reach their maximum range. For example, the "ok" gesture only requires a slight pinching together of the index finger and thumb, which involves limited movement of each finger, whereas the "fist" gesture requires all five fingers to move together, nearly reaching the maximum range of movement for each finger. If the search space for regression is (much) larger than necessary, implying too many potential outcomes, it will make accurate regression more difficult [35], leading to increased estimation error. Therefore, our main idea is to dynamically narrow down the search space of each time window to an appropriate range before performing the regression, which we find in Section IV can further reduce the tracking error effectively.

In principle, this idea applies for all three branches. But to avoid unnecessary system overhead as described above, we only apply it to the second branch in the current WETrak.

Solution: To narrow down the regression search space of EMG data for each input window, our main idea is to estimate the actual range of EMG data in the current input window (i.e., estimate its actual maximum and minimum values), and use this range (or a slightly expanded one) to perform regression. To convert this idea to a practical solution, we divide the theoretical ranges of each DoF (from existing kinematics study [34]) into bins. We then cast the problem of finding the actual DoF range as two

[4]We find that adding weights to the features extracted by the encoder is more effective than adding them directly to the input EMG signals.

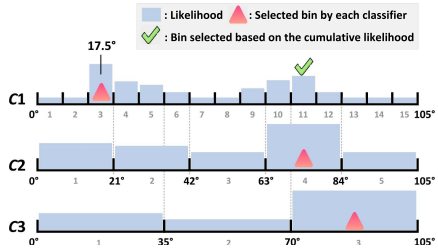

Fig. 7. Illustration of the branch adapter design.

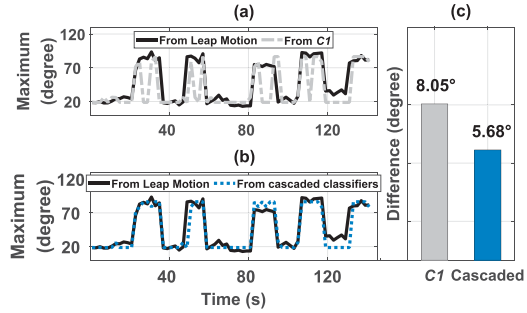

Fig. 8. Differences of the actual maximum PIP's DoF of a middle finger between Leap Motion and (a) using $C1$ or (b) using cascaded classifiers. (c) The overall differences across all DoFs of the middle finger.

classification problems — the bins to which the actual maximum and minimum DoF values belong.

The procedure for estimating these two boundary values is similar, and we use the maximum value as an example to demonstrate our design. The second branch outputs 12 DoFs, and we divide the theoretical range of each DoF into R bins (classes), where the setting of R will be discussed below. For each input window of EMG data $\{F_t^{in}\}_{t=1}^{s}$, we design a lightweight classifier consisting of six convolution layers to predict 12 values, where each value expressed as c_k^{max} is the maximum value of the kth DoF. Thus, the classifier $h(\cdot)$ learns the following relation:

$$h(\{F_t^{in}\}_{t=1}^{s}) \rightarrow C = \{c_k^{max}\}_{k=1}^{12}, \tag{4}$$

where we can use the center value of bin c_k^{max} as the maximum value of the kth DoF. For example, for the classifier $C1$ in Fig. 7, the DoF is divided into 15 bins. Assuming the theoretical range of this DoF is 0–105°, each bin spans 7°. The bars drawn for each bin represent the likelihood of the classification result, and the third bin with the highest likelihood will be selected. In this case, this c_k^{max} is 17.5° (=7° + 7° + $(\frac{7}{2})$°).

In fact, this classifier can be directly extended to estimate the 12 minimum DoFs simultaneously by adding another 12 outputs, i.e., the classifier outputs a total of 24 classification results, where the first 12 results correspond to the 12 minimum DoFs and the rest corresponds to the 12 maximum DoFs. In this solution, the performance of the classifier is critical. Therefore, we propose the following mechanism to further improve the classification performance.

1) Cascaded classifiers: This mechanism is related to the setting of the bin number R of the classifier. When R is larger, the span of each bin is smaller. In this case, the estimated maximum or minimum bounds are more accurate if the bins are chosen correctly. However, when the number of output classes is large, we find that it becomes more difficult to find accurate classification results directly for EMG data. Conversely, if R is small, it is easier to find the bins correctly, but the span of each bin is too large, which also leads to less accurate estimated boundary values. To address this issue, we propose a mechanism to cascade multiple classifiers with different "resolutions". We first use classifiers with smaller R to more accurately locate the rough range of the correct result. We then switch to higher resolution classifiers to gradually narrow their range and accumulate the likelihood results from each classifier. Finally, we will select the bin with the highest resolution from the classifier with the largest R, based on the highest *cumulative* likelihood of all classifiers as the final result.

Fig. 7 shows an example with three classifiers $C1$–$C3$, where $C1$ has the highest resolution with 15 bins, $C2$ has the intermediate resolution with 5 bins, and $C3$ has a minimum resolution of 3 bins. Fig. 7 is illustrated based on a real example, where the actual maximum boundary value is located in the 11th bin of $C1$. The classification results of $C2$ and $C3$ both provide correct classification results. For $C1$'s own output, the highest likelihood is in the third bin, and the correct (11th) bin also has a relatively high likelihood. For each bin of $C1$, we make the final choice by selecting the bin with the highest cumulative likelihood among all three classifiers, which compensates for a single misclassification by the highest resolution classifier.

In the current WETrak, we use the cascaded classifier for each DoF of the second branch, and find that it can effectively improve classification reliability compared with using only one classifier. Fig. 8(a) shows a trace for the PIP's DoF of a middle finger. The middle finger is repeatedly extended and flexed, and we compare the actual DoF maximum (obtained from Leap Motion) in each input window with the estimated result using only $C1$. We can see that large differences often occur, especially when the fingers are flexed. Fig. 8(c) shows that the median difference is about 8.05°. When cascaded classifiers are used, the classification results are more accurate, as shown in Fig. 8(b), which can reduce the difference by 29.44% to 5.68°.

We note that this mechanism is not designed to completely avoid classification errors. As long as it can reduce the classification error, we find that it can effectively improve the final tracking performance. This mechanism can be further enhanced by slightly widening the estimation range to tolerate some classification errors, and we will explore such opportunity in the future. As shown in the evaluation, the current design of this mechanism can bring another 24% tracking error reduction.

2) Loss function: To train each classifier, we design three loss functions. The first L_{ce} penalizes misclassified results:

$$L_{ce} = -\sum_r p_r^k \times \log\left(p_r^k\right), \tag{5}$$

where p_r^k represents the likelihood of the kth DoF being classified into the rth bin. However, since our goal is to identify the bins containing the minimum and maximum values of the DoF, uniformly penalizing all misclassified bins cannot distinguish the degree of their inaccuracy. In WETrak, the greater the

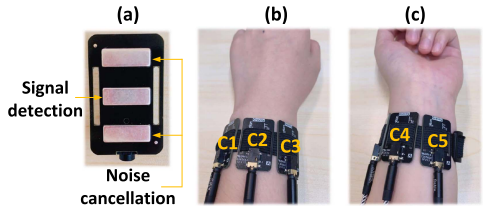

Fig. 9. Illustration of (a) the EMG sensor used in WETrak and (b) and (c) the front and back of our wristband prototype with five EMG sensors C1–C5.

distance between the predicted bin and the actual bin, the worse the error. Therefore, we further introduce two *distance-related* loss functions to guide the classifier.

The second loss L_{itr} quantifies an *inter-class distance*, which can impose a larger penalty on misclassified bins that are farther away, guiding the classifier's estimate as close as possible to the correct class:

$$L_{itr} = \sum_{r} \left(\hat{r}^k - r^k\right)^2, \tag{6}$$

where $\hat{r}^k$ is the subscript of the correct bin associated with the kth DoF from Leap Motion, and r^k is the subscript of the estimated bin associated with the kth DoF from the classifier, which can be obtained by selecting the one with the largest likelihood p_r^k. Finally, we consider an *intra-class distance*, i.e., samples within the same class should be distributed as close as possible. So, we introduce L_{ita} to calculate the difference between the features f^k (where f^k is a feature vector) of the samples in the class and the corresponding class centers c^k:

$$L_{ita} = \sum_{r} \left(f^k - c^k\right)^2, \tag{7}$$

where c^k is a learnable parameter. We then minimize the overall loss $L_{ce} + L_{itr} + L_{ita}$ during classifier training. The classifier is trained before the tracker network.

D. Adaptation to New Users

Even for the same hand pose, there are differences in the EMG signals performed by different users [36]. Therefore, a trained tracker is usually not directly used for a new user (whose data does not participate in training) [15], [16], [37]. To minimize the overhead of model adaptation, we use fine-tuning to adapt the tracker, rather than training from scratch. Specifically, for any tracker trained before or published in the community (we will lead this work in the future), we collect only a small amount of EMG data (and their labels) for new users to fine-tune the model, e.g., 10% of the data volume used for training the tracker from scratch. In the future, we will study how to further avoid collecting labels when fine-tuning.

IV. Performance Evaluation

Testbed: We develop WETrak using the OYMotion EMG sensor [38], as shown in Fig. 9(a). Each WETrak wristband consists of five EMG sensors, denoted as C1–C5 in Fig. 9(b)-(c), strung together by a nylon elastic strap. The default sampling rate of the EMG sensor is 500 Hz, and the input window size is empirically set to 2 seconds for 1000 input frames. As a proof-of-concept prototype, we are using wires to collect EMG signals in the current development, and we will replace them with Bluetooth in the future. To synchronize the multi-channel EMG sensors, we optimize the code execution time as much as possible at the software level to ensure there are no unnecessary delays or computations between readings from each sensor. We implement the neural network using PyTorch 1.8 and train it on a GeForce RTX 2080Ti GPU using gradient descent based on the Adam optimizer with a learning rate of 0.01 and a batch size of 16. To prevent overfitting, we use Dropout with a dropout probability of 0.05 during training. After the model is trained, we deploy it to a Google Pixel 7 phone for inference using TFLite [39]. It can output 500 frames per second of 20 DoFs, which can be used to reconstruct hand poses of 15 skeleton points (Section II-B).

Data collection: We invite 15 users to participate in our data collection. This study received ethical approval from the University's Human Subjects Ethics Sub-Committee, application number H002205 for human ethics review. These users range from regular exercisers to those who do not engage in any form of physical activity, and vary in weight from 45–92 kg. We clean the skin around the user's wrist and then they can wear our wristband in the positions they would normally wear their smartwatches. The C2 sensor acts as an anchor point for wearing WETrak, centered on the front side of the user's wrist in Fig. 9(b). Once the wristband is worn by the user, we adjust the sensors' positions to ensure they are in the optimal locations.

Since the potential range of hand motions encompasses a set of base states [40], we collect a dataset covering all the base states to comprehensively evaluate WETrak. Specifically, there are a total of 22 basic states as shown in Fig. 11, including all movements of 1–5 fingers. Users are required to perform these 22 basic states sequentially, each for one minute. During the actual use of the system, since the wearing position of WETrak cannot be exactly the same every time, we take this factor into consideration, and each user takes off the wristband at least once and puts it on again to continue data collection. Throughout the data collection process, we use Leap Motion [33] to collect ground truth data for training and evaluation purposes.

Training: We divide 15 users into two groups of 10 "seen users" and 5 "unseen users". For each seen user, we utilize 80% of the data to train a model individually and the rest for evaluation. For unseen users, instead of training a new model from scratch, we fine-tune the model already trained on the seen users, which allows WETrak to quickly adapt to new users. For unseen users, we evaluate their tracking performance using different amounts of fine-tuning data, and the default amount is 10% of the data collected from each unseen user.

Metric: We compute the absolute error in degrees for each finger skeleton point along all DoFs unless otherwise stated.

A. Overall Performance

In this subsection, we compare the performance of the following three settings in the evaluation:

- *NeuroPose [16] with wristband:* NeuroPose is the state-of-the-art finger tracking system with EMG signals from

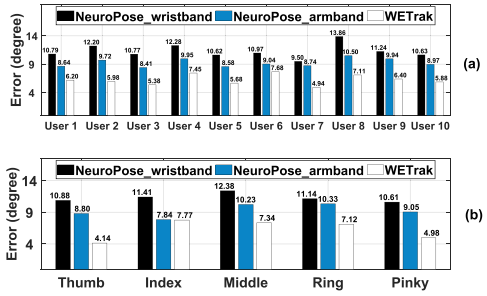

Fig. 10. Tracking errors for (a) different users and (b) different fingers.

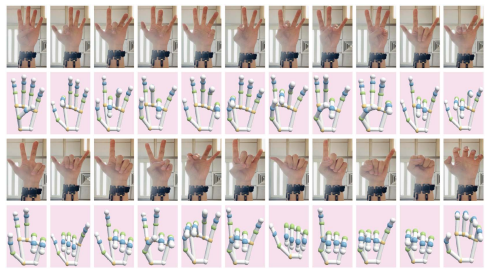

Fig. 11. Illustration of all basic states of finger movements from the camera and our WETrak, respectively.

Myo [17], a commercial armband worn on the forearm, to reconstruct finger movements. First, we directly employ the NeuroPose tracking method to the captured EMG signals using our wristband.

- *NeuroPose [16] with armband:* Then we employ the NeuroPose tracking method to signals from its original armband, i.e., Myo, worn on the forearm.
- *WETrak:* Last, we employ our proposed finger tracking system to signals from our wristband.

Accuracy: In Fig. 10(a), it can be seen that if we directly apply the NeuroPose tracking method to the wristband, its tracking error is large across different users, e.g., 9.50–13.86°. For a fair comparison, we further asked each user to repeat our data collection while wearing the Myo armband to obtain the same amount of EMG data. In this setting, NeuroPose with armband for the ten seen users has median tracking errors from 8.41° to 10.50° across all 15 skeleton points (we observe that the error of NeuroPose with armband is slightly higher than its original paper's 6.24° on our dataset that fully covers all basic states of hand motion, and to facilitate future research, we will open-source our code for easy comparison). As shown in the ablation study to be presented (Fig. 12(a)), with efficient sensor placement and network design, WETrak overcomes the suboptimal location problem at the wrist for finger tracking and achieves comparable performance to NeuroPose with armband, while the full version WETrak of branch adaptation finally reduces the error to 4.94–7.68°, about 32.22% lower than NeuroPose with armband.

Fig. 10(b) shows the tracking performance of the three settings across different fingers. Similarly, the tracking error of NeuroPose with wristband is large, e.g., 10.61–12.38°. For a fair comparison, NeuroPose with its original armband has a median error range of 7.84° to 10.33°, while WETrak reduces it to a range of 4.14° to 7.77°. Since the thumb and the pinky have their individual network branches in WETrak, they have slightly less errors than other three fingers.

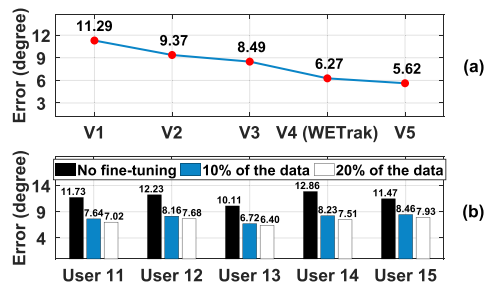

Fig. 12. (a) Ablation study. (b) Tracking performance for new users with varying amounts of data used for fine-tuning.

Visualization: We further visualize the tracking results of WETrak. We develop a tool to visualize a hand of 15 finger skeleton points based on 20 DoFs. Fig. 11 shows that the hands reconstructed by WETrak match the actual hands well.

B. Micro-Benchmarks

Ablation study: We first conduct an ablation study to investigate the efficacy of each technique module in WETrak. So we develop the following intermediate WETrak versions:

V1: it uses only one model branch to track all fingers, similar to NeuroPose. However, due to the more dense and distorted distribution of muscles at the wrist, one branch has poor tracking ability, with a median error of 11.29° in Fig. 12(a).

V2: it employs three branches based on our study of muscle distribution in Section III-A. Each branch only takes the EMG signals of the most relevant channels as input in *V2*, which can reduce the tracking error to 9.37° in Fig. 12(a).

V3: it retains the three branches of *V2*, but takes EMG signals from all channels through attention across all channels. Fig. 12(a) shows that the error can be further reduced to 8.49°. In summary, based on our study of muscle distribution, effective sensor placements at the wrist and specialized model design, errors are reduced from 11.29° (in *V1*) to 8.49° (in *V3*) with 25% reduction. This error is already comparable to that of NeuroPose using an armband with better EMG placement.

V4: it is the full version WETrak, which adds the regression adaptation design to assist the tracking of the index, middle and ring fingers, with another 26% error reduction in Fig. 12(a).

V5: it employs five different branches, one for each finger. As each branch is dedicated to one finger, *V5* should actually get better tracking performance. For example, it has a median error of 5.62Â°, slightly better than *V4*. However, since WETrak is positioned to be integrated into wearables in the future, its calculation will be performed by phones or even directly by wearables. So, we prefer to take as few branches as possible to avoid unnecessary energy and memory overhead, but we will do further improve the performance of WETrak in the future.

New users: Fig. 12(b) shows the tracking performance of WETrak on five new users. If we apply the models of all seen users to new unseen users without model fine-tuning, the average

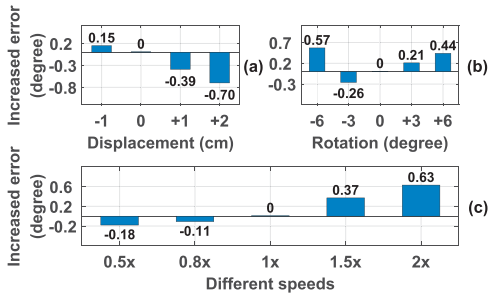

Fig. 13. Tracking performance at (a) different device displacements, (b) different device rotations, and (c) different finger movement speeds.

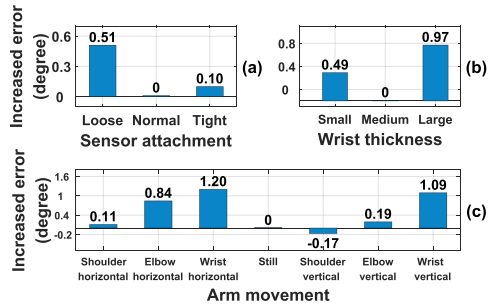

Fig. 14. Tracking performance at (a) different sensor attachments, (b) different wrist thicknesses, and (c) different arm movements.

error is 10.11–12.86°, which is relatively high. With fine-tuning (Section III-D), even with only 10% of the data collected from each unseen user (i.e., 2.2 minutes of data), the tracking error is reduced to 6.72–8.46° with the reduction of 26.24–36.00%. When fine-tuned with 20% of the data, the error of 7.31° is comparable to that of seen users with the model trained from scratch, suggesting WETrak can quickly adapt to new users.

Effect of wearing position: When the user wears WETrak, we use the C2 sensor as an anchor point, which should be placed in the center on the front of the user's wrist. However, after the user wears WETrak, the wearing positions cannot be exactly the same each time. Thus, in this experiment, we study the effect of wearing position in two aspects: the displacement of the device along arm and the rotation along the wrist.

We first investigate the effect of displacement. Starting from the position where the user normally wears the smartwatch (the *baseline position*), we denote the displacement towards fingers as (-) and towards the elbow as (+). In Fig. 13(a), we change the wearing position from −1 cm to +2 cm (since it is too loose to wear the device at −2 cm, we start at −1 cm). Fig. 13(a) shows the difference in median error at each displacement relative to the baseline position. We can see that the error increases slightly by 0.15° when the band is closer to fingers, and decreases when the band is moved towards the elbow. This is because muscle distribution becomes clearer along this direction, resulting in better EMG signals.

Then we examine the effect of rotation. Starting from the baseline position, we rotate the wristband clockwise (+) and counterclockwise (-) at angles of 3 and 6°, and examine the error under each angle. Fig. 13(b) shows that WETrak can tolerate such moderate rotations that occur in practice, with additional errors varying from −0.26° to 0.57°.

Effect of finger movement speed: This experiment examines the tracking performance at different finger movement speeds. The baseline speed is the natural movement speed, e.g., it takes about 1.25 seconds for the hand to fully extend from the fist. In this experiment, users intentionally decrease the movement speed to 0.5x and 0.8x, and increase the speed to 1.5x and 2x. In Fig. 13(c), the error increases less than 1° in all cases, indicating that the movement speed has a small influence on WETrak. The error decreases slightly when the movement speed is slower, probably due to the greater force exerted by muscles controlling the slow movement.

Effect of sensor attachment: In this experiment, we investigate the effect of the sensor attachments on tracking performance. Initially, users are asked to adjust the wristband to a level of tightness they find comfortable, which is referred to as the "Normal" state. Subsequently, the length of the wristband is either loosened or tightened by 1 cm, denoted as "Loose" or "Tight" states, respectively. As shown in Fig. 14(a), the tracking error of "Loose" state increases significantly, e.g., 0.51°. This can be attributed to insufficient contact between the EMG sensors and the skin, which leads to weakened signals and the additional noise. The tracking error in the "Tight" state is higher by 0.1° compared to the "Normal" state. This is likely due to the fact that an overly tightened wristband can inhibit natural muscle movement, thereby degrading the quality of the EMG signals.

Effect of variations in wrist thickness: In this experiment, we investigate the effect of wrist thickness on tracking performance. According to [41], we know that the wrist size is related to gender and height. Based on these rules, we measure the wrist sizes of fifteen users and categorize them into three groups: "Small", "Medium" and "Large". We then calculate the average tracking error for each group, and the results are depicted in Fig. 14(b). It is observed that individuals with either small or large wrists experience an increase in tracking error of 0.49° and 0.97°, respectively. The observed increase in tracking error for the small and large wrist size groups may be related to variations in muscle density or distribution, which can influence the accuracy of EMG signal detection.

Effect of arm movement: In this experiment, we investigate the effect of arm movement on finger tracking performance. The arm consists of three skeletal joints: the shoulder, elbow, and wrist. Therefore, we categorize arm movements into three types: shoulder-only movement, elbow-only movement, and wrist-only movement. Each type of movement is further divided into horizontal and vertical directions. The results are shown in Fig. 14(c). "Still" denotes the condition where only finger movements occur with the arm remaining stationary. It can be observed that wrist movements have a more significant impact on finger tracking, with errors increasing by 1.20° and 1.09°, respectively. This is likely because EMG sensors are positioned on the wrist, where wrist movements can cause friction between the sensors and the skin surface, introducing additional noise. Given the distance between the elbow and the wrist, the impact of elbow movement on finger tracking performance is relatively

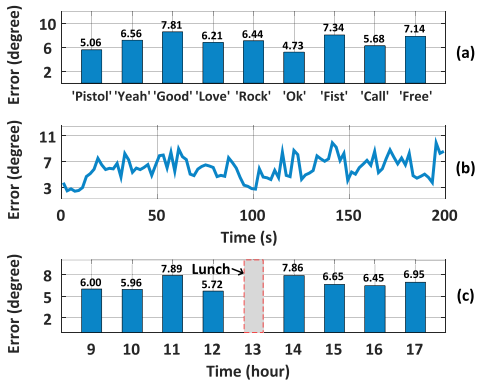

Fig. 15. (a) Tracking error over common hand gestures. (b) Tracking error does not accumulate over time. (c) Tracking performance during an 8-hour working period.

minor, with errors increasing by 0.84° and 0.19°, respectively. The shoulder, being the furthest from the wrist, has a negligible effect on finger tracking performance.

Tracking error over common hand gestures: In this experiment, we investigate the tracking error over common hand gestures in daily life. These gestures include eight specific movements: pistol, yeah, good, love, rock, ok, fist, and call. Additionally, there is allowance for free finger movement without any constraint. The tracking performance across these nine scenarios is illustrated in Fig. 15(a). It can be observed that tracking errors range from 4.73° to 7.81°, with an average error of 6.33°, indicating the performance of WETrak is relatively stable over different hand gestures.

Tracking error over time: In this experiment, we examine whether tracking errors for WETrak accumulates. Fig. 15(b) plots the results over 200 seconds, where the error varies around 6° and is not cumulative. This is because each window of input sensing signals is processed independently of previous or future windows, preventing errors from accumulating.

Long-time performance: We further investigate the tracking performance of WETrak over a long period of time. Specifically, a user wears our wristband throughout working hours, 9 am–5 pm, for daily activities such as typing, drinking water, making phone calls, etc. To evaluate performance, the user performs finger movements in front of Leap Motion for three minutes at the end of each hour. If the user needs to leave the desk (due to the wired connection in our current prototype), the user can remove the wristband and wear it again after coming back. Fig. 15(c) shows that the tracking performance is relatively stable in different time periods and suitable for long-time use.

C. System Overhead.

We examine the system overhead on a Google Pixel 7 phone.

Memory: With our compact network branch design, WETrak has a runtime memory of 49.6 MB, including all input/output tensors, intermediate computation results, and necessary caches. This ensures that WETrak can be resource-friendly for integration into wearables and executed on mobile phones or even directly on wearable devices in the future.

TABLE I
ENERGY CONSUMPTION OF WETRAK

Sliding Step	Energy consumption	Discharge rate of battery
2.0 s	131.09 mAh	3.01%
1.0 s	144.59 mAh	3.32%
0.5 s	165.05 mAh	3.79%
0.1 s	376.27 mAh	8.64%

Energy and latency: We measure the energy consumption of WETrak using the Batterystats and Battery Historian tools from Android [42]. When using the default window size of the model input, e.g., 2 seconds in the current WETrak, we change the step to slide the window from 2 to 0.1 seconds. When the sliding step is 2 seconds, two consecutive windows do not overlap. When the sliding step is 1 s, two consecutive windows have an overlap of 1 s. Smaller sliding steps result in more timely tracking output. When the windows overlap after sliding, we can simply discard the results that have already been output in the previous window, and only output the results of the non-overlapping part in the new window.

For each 2-second input window, WETrak can finish all computations and output tracking results within 0.039 seconds. So even with a sliding step of 0.1 seconds, WETrak is still lightweight enough to operate in time. On the other hand, WETrak is also energy efficient as shown in Table I. When sliding step is not less than 0.5 seconds, the power consumption is 131.09–165.05 mAh, and the maximum discharge is 3.79% per hour. Even performing tracking every 0.1 seconds, the energy consumption increases to 376.27 mAh, resulting in a discharge rate of 8.64% only. In practice, a sliding step of 1-2 seconds can achieve a better trade-off between energy efficiency and output timeliness.

V. RELATED WORK

Contactless finger tracking: Significant advances have been made in the field of finger tracking over the years. Early contactless finger tracking approaches typically relied on vision-based methods, which are primarily divided into three categories, including optical motion capture systems [43], [44], depth sensing systems [11], [33] and RGB camera based systems [9], [10]. They used cameras and computer vision algorithms to analyze frames of finger movement to determine the position and orientation of each finger. However, they often have privacy concerns and require a clear line of sight between the camera and hand, and can be affected by changes in lighting conditions [45]. Furthermore, they usually require significant computing resources and power, which limits their use in mobile or low-power devices.

To solve these problems, wireless-based methods have been proposed recently. They utilize various wireless signals such as millimeter wave [12], [13] and Wi-Fi [14], [46]. These methods will work in any lighting condition and may not require line of sight. However, they can be susceptible to interference and depend on the sensing environment. Furthermore, these methods mainly work within relatively limited service coverage [15].

Wearable-based finger tracking: Previously, numerous studies focused on hand gesture recognition using wearable

devices, such as Tomo [47], Aurasense [48] and Pyro [49], which primarily deal with distinguishing between pre-defined hand gestures [50]. Unlike these classification tasks, which identify specific, static hand configurations, finger tracking involves continuously predicting the movements of the fingers. This is a more general and challenging regression task that requires capturing finer, dynamic details of finger movements. Our system, WETrak, is specifically designed to address this task, thereby offering a more flexible and comprehensive solution for applications that require detailed real-time interaction, such as advanced virtual reality environments. To achieve this goal, a variety of works have been proposed. For example, glove-based or ring-based methods [51], [52], [53], [54], [55], [56] typically use inertial measurement unit (IMU) sensors, magnetic sensors or light sensors embedded in a glove or multiple rings to measure finger joint angles. However, they can be invasive and interfere with daily activities involving the fingers. Additionally, IMU sensors are prone to accumulating errors over time. In addition to such designs, recent work has begun to explore the use of wrist-worn devices for tracking such as using thermal sensors [57] and using antenna impedance characteristics [58]. However, thermal sensors are easily affected by surrounding heat sources. Antenna impedance can be influenced by other radio frequency signals and/or the environmental factors.

Recently, EMG sensors, which measure the electrical activity generated by skeletal muscles, have been increasingly used for finger tracking [16], [59], i.e., the locations of hand skeletons with respect to the wrist coordinate system. This is because they capture the muscle activity associated with finger movements. Some works, like [15], have shown that by combining IMU data with EMG data, it is possible to track the position of the entire arm in 3D space. However, these works rely on the armband [17], a device that needs to be worn on the forearm only for tracking purpose, which makes the user's daily activities inconvenient. Therefore, WETrak advances the possibility of future integration into wrist-worn wearables by solving new challenges to deploy EMG from the forearm to the wrist, focusing on tracking finger movements relative to the wrist. Our WETrak is orthogonal to the task of using IMU data at the wrist to determine the wrist's location [60], [61]. In the future, we plan to integrate them.

VI. Conclusion

This paper presents WETrak, a finger tracking system using wrist-worn EMG sensors. State-of-the-art EMG-based finger tracking methods primarily use armbands, which require the user to purchase a special armband only for tracking purpose and wear it on forearm each time. This overhead has hindered widespread adoption of finger tracking in practice. This paper demonstrates the feasibility of moving EMG sensors from the forearm to the wrist for finger tracking by solving two new challenges, which brings a strong incentive for future integration of EMG sensors into existing wrist-worn wearables. We developed a prototype system of WETrak, and extensive results show that WETrak has achieved valuable performance gains over the state-of-the-art method.

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